



Assessment Analysis Guide
25-26 Math G6 Form B Write/Comp Ratios
Mathematics
Grade 6

Reference #: 682f71a

1

Item Id: i184941

Item Type: SR (Selected Response)

Standards Description:

6.RP.A.3.a

Make tables of equivalent ratios relating quantities with whole-number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane. Use tables to compare ratios.

Content:

The table below represents equivalent ratios.

8	10
12	15
24	30
m	40

What is the value of m in the table?

- A28
- B32
- C34
- D36

Primary Standard: 6.RP.A.3.a

Points Possible: 1

Distractor Rationale:

- A:28

Student used just the column with the missing value to identify a pattern based on multiples and incorrectly extended it, rather than considering the proportional relationship between corresponding values.
- B:32

Correct.
- C:34

Student applied an additive pattern from the previous row to determine the missing value, rather than recognizing the multiplicative relationship that maintains the proportionality in the table.
- D:36

Student used just the column with the missing value to identify an additive pattern between two consecutive values and incorrectly extended it to determine the

missing value, instead of applying a multiplicative relationship to maintain proportionality.

2

Item Id: i185010

Item Type: SR (Selected Response)

Standards Description:

6.RP.A.1 Understand the concept of a ratio and use ratio language to describe a ratio relationship between two quantities.

Content:

A committee of teachers and students is meeting with the school board to request funding for extracurricular activities. The committee is creating posters for the meeting using a box of markers. The box contains 28 black markers and 16 red markers.



What is the ratio of black markers to red markers?

- A 4 : 7
- B 4 : 11
- C 7 : 4
- D 7 : 11

Primary Standard: 6.RP.A.1**Points Possible:** 1

Distractor Rationale:

A: 4 : 7

Student confused the roles of the given values in setting up a ratio and found the reverse ratio.

B: 4 : 11

Student interpreted the context as requiring a ratio of part to whole instead of part to part.

C: 7 : 4**Correct.**

D: 7 : 11

Student interpreted the context as requiring a ratio of part to whole instead of part to part.

3

Item Id: i184938

Item Type: SR (Selected Response)

Standards Description:

6.RP.A.3.a

Make tables of equivalent ratios relating quantities with whole-number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane. Use tables to compare ratios.

Content:

The tables below show ratios of teaspoons of sugar to ounces of iced tea. Each table represents a different recipe.

Which table represents the recipe with the greatest amount of sugar, in teaspoons, per ounce of iced tea?

A

Sugar (tsp.)	Iced Tea (oz.)
6	24
12	48
15	60

B

Sugar (tsp.)	Iced Tea (oz.)
3	18
4	24
6	36

C

Sugar (tsp.)	Iced Tea (oz.)
1	3
5	15
8	24

D

Sugar (tsp.)	Iced Tea (oz.)
7	24
14	48
28	96

Distractor Rationale:**A: Sugar (tsp.) Iced Tea (oz.)**

6	24
12	48
15	60

Student applied an additive reasoning approach instead of a multiplicative one, comparing only the first row of each table and selecting the greatest additive difference, OR student compared tables based on a single data point rather than analyzing the proportional relationship.

B: Sugar (tsp.) Iced Tea (oz.)

3	18
4	24
6	36

Student correctly calculated unit rates but confused the order of quantities in the ratio, selecting the greatest value without considering the intended relationship between the two quantities, OR student correctly calculated unit rates with the correct order of quantities in the ratio, but then confused the least with the greatest.

C: Sugar (tsp.) Iced Tea (oz.)

1	3
5	15
8	24

Correct.**D: Sugar (tsp.) Iced Tea (oz.)**

7	24
14	48
28	96

Student thought that the table with the largest values of both quantities represented the greatest ratio, rather than analyzing the proportional relationship between the two quantities.

4

Item Id: i184940

Item Type: SR (Selected Response)

Standards Description:

6.R.P.A.1

Understand the concept of a ratio and use ratio language to describe a ratio relationship between two quantities.

Content:

In the 7th grade, there are 84 students who are either gymnasts or hockey players. 24 of the students are gymnasts, and the rest are hockey players.

Which statement is true?

- A

For every 5 hockey players, there are 2 gymnasts.
- B

For every 5 gymnasts, there are 2 hockey players.
- C

For every 7 hockey players, there are 2 gymnasts.
- D

For every 7 gymnasts, there are 5 hockey players.

Primary Standard: 6.R.P.A.1

Points Possible: 1

Distractor Rationale:

- | | | |
|----|---|--|
| A: | For every 5 hockey players, there are 2 gymnasts. | Correct. |
| B: | For every 5 gymnasts, there are 2 hockey players. | Student reversed the order of the quantities in the ratio. |
| C: | For every 7 hockey players, there are 2 gymnasts. | Student incorrectly used the total of the quantities as one part of the ratio and then correctly simplified an incorrect ratio. |
| D: | For every 7 gymnasts, there are 5 hockey players. | Student incorrectly used the total of the quantities as one part of the ratio, but determined the other quantity correctly and then correctly simplified an incorrect ratio. |